

Orli - A new take on Print-on-Demand

Project Team

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Chapter 1

Introduction

This scope document outlines the proposed development of a digital platform designed to empower local artists by connecting them with global markets through stream-lined, AI-enhanced e-commerce experience. The purpose of this document is to define our vision, the boundaries, and the technical direction of the project, ensuring clarity for all stakeholders and establishing a shared understanding of what the system will—and will not—deliver.

The primary aim of this document is to present a clear and actionable road-map for building said web-based platform that may serve as a foundational reference for the project team, academic evaluators, and potential collaborators. It may detail the system's intended features, technological stack, development phases, and limitations. By articulating the scope now, we aim to prevent scope creep, aligning expectations, and guide decision-making throughout the life cycle of the project.

In today's digital economy, platforms like Shopify, Etsy, WooCommerce, Printify, and Gelato have revolutionized how creators monetize their work. However, these systems often cater to established businesses or lack localized support for emerging artists or business-minded individuals. This project seeks to bridge that gap by offering a tailored solution for student creators, independent designers, and amateurs, particularly those with limited resources. The envisioned platform will be built using the MERN stack—MongoDB, Express.js, React, and Node.js—providing a robust and scalable foundation for web development. AI modules, developed in Python, will offer intelligent design assistance and personalization.

This document sets the stage for a collaborative, well-defined development process—one that celebrates creativity, leverages technology, and delivers real-world impact for emerging artists and entrepreneurs.

1.1 Problem Statement

In the evolving landscape of the print-on-demand industry, there is a notable absence of an integrated platform that bridges the gap between local talent and global markets. Existing solutions fall short in offering advanced AI-powered design tools and a marketplace that caters to both local and international demands. This gap creates challenges for local artists to showcase their work on a global stage and limits customers' access to unique and culturally rich designs. It's a missed opportunity for both creators and consumers.

Expanding further, Pakistan's creative sector lacks a streamlined print-on-demand (POD) infrastructure, leading to inefficiencies in production, higher costs, and slower delivery times for individuals interested in personalized or unique products. This gap limits the growth of independent designers and weakens the customer experience. Our platform addresses this by introducing a localized POD model tailored for emerging artists and entrepreneurs. Beyond simplifying production and fulfillment, it empowers creators to build and manage their own brands—laying the foundation for sustainable digital entrepreneurship in Pakistan.

1.2 Motivation

We aim to empower local artists, streamline production and offer customers access to a diverse array of unique designs. The motivation behind this project stems from the lack of accessible, integrated print-on-demand platforms in Pakistan—especially for student creators and independent designers. Existing global solutions like Printify and Gelato are not optimized for local infrastructure, leading to inefficiencies in production, higher costs, and limited market reach. This project aims to bridge that gap by offering a localized, AI-enhanced platform that empowers artists to launch and manage their own micro-brands without going through all the pains and risks that any new business has to go through. By combining intuitive design tools, streamlined order fulfillment, and scalable technology, we hope to foster creative entrepreneurship and improve customer experience across the local market.

1.3 Problem Solution

The project caters to our problem statement using an integrated platform that leverages advanced AI-driven design tools and connects creators with global markets. It aims to foster local talent, invigorate local economy and enrich the cultural tapestry of the global marketplace. It would also empower creators to reach wider audiences and provide con-

sumers with exclusive access to diverse meaningful designs. The proposed platform will utilize the MERN stack—MongoDB, Express.js, React, and Node.js—providing a robust and scalable foundation for web development. The AI modules will offer seamless design experience to artists. Our goals and objectives are as follows:

1. expanding local artists' visibility,
2. expanding market reach, and
3. fostering cultural exchange

Our goals are to:

- Create an intuitive platform,
- Achieve high user engagement,
- Allow people to create sustainable businesses with ease

Our platform would function as an all-inclusive hub combining the user-friendly design of e commerce platforms like Shopify with the customization capabilities of Printify or Gelato. We will emphasize ease of use for both creators and customers, stream-line order fulfillment and a strong community aspect connecting artists indirectly with their audience.

1.4 Stakeholders

These kind of projects are generally all-encompassing and have a range of stakeholders. In our case, our most prioritized stakeholders would include:

1. The Academic Evaluators
2. Our Team
3. Manufacturers
4. Artists/Designers
5. Competitors like Printify, gelato, etc. (if we go Global)

Chapter 2

Project Description

This section details the specifics and finer points of the software project, expanding on what the features entail and how they will be accomplished for the final delivery of the product.

2.1 Scope

By bridging the gap between manufacturers, creative designers, and consumers, our proposed platform fosters a collaborative and decentralized approach to product creation, customization, and commerce. It will serve as a comprehensive digital ecosystem for artists and designers, offering a range of core functionalities tailored to support creative entrepreneurship and localized print-on-demand services. Designers/Users will be able to create personalized profiles, upload original designs, and utilize AI-assisted tools for design generation and enhancement. This would be powered by three AI models each for design generation, enhancement, and mock up context generation. The system will include product customization features in terms of designs with real-time previews, enabling creators to visualize their work on various merchandise formats. A dedicated manufacturer dashboard will facilitate order tracking and production coordination, while integrated analytics tools will provide insights into sales performance and customer engagement. The platform will also host a curated marketplace for browsing and purchasing the designed products, supported by SEO-optimized e-commerce capabilities to enhance discoverability. Importantly, the scope excludes custom plugin development, advanced marketing automation, and enterprise-level integrations—ensuring the solution remains focused, technically feasible, and aligned with the team’s current resources and expertise. These exclusions are intentional, ensuring the project remains focused, achievable, and aligned with the team’s current capabilities and resources.

2.2 Modules

In its most rudimentary form, our app is an e-commerce platform. But on a closer look, it's a web-application that re-imagines the traditional online marketplace by merging the core functionalities of platforms like Printify, Shopify, and Etsy into a unified ecosystem. Unlike conventional e-commerce models in Pakistan, our platform serves as a dynamic, multi-sided marketplace that connects three key stakeholders: manufacturers, independent designers, and end consumers. At its core, the platform empowers designers to build their own brands and storefronts by sourcing materials directly from verified print providers. Designers can create and showcase custom products using integrated design tools, including MS Paint-style interfaces and AI-powered modeling features. These tools allow for real-time 3D rendering enabling customers to visualize products on themselves or in their environment before making a purchase. Below is the architectural visualization of our system focusing on each module separately to ensure all the details are up-front.

2.2.1 Module 1: User Experience (React)

This module handles all user-facing interactions, tailored to specific roles. Each UI is a self-contained module with its own routing, state management, and API bindings. It is subdivided into sub-modules pertaining to each perspective, customer, designer, manufacturer, and admin

2.2.1.1 Customer UI

1. Preview
2. Checkout and Track Orders
3. View Trends
4. Browse Products

2.2.1.2 Designer UI

1. Design Canvas (Adobe-like features)
2. Storefront Creation and Management
3. Bulk Order Checkout
4. Accept Orders

5. Trend Analytics
6. Product Page SEO Optimization for Storefront

2.2.1.3 Manufacturer UI

1. Accept Orders
2. Print and Ship
3. Inventory Management
4. Performance Statistics

2.2.1.4 Admin UI

1. User Management
2. System Monitoring
3. Content Moderation
4. Analytics Dashboard

2.2.2 Module 2: Core Business Logic (Node.js + Express Backend)

This module acts as the orchestrator between frontend, database, and external services. Each service is modular and can be containerized or scaled independently. It is divided into the following features or sub-modules:

1. Product Service
2. Order Service:
3. Payment Service
4. Auth Service
5. Messaging Service
6. SEO Service

2.2.3 Module 3: AI and Creative Intelligence (Flask + TensorFlow)

This module is dedicated to enhancing user creativity and product visualization. This module can be deployed separately for scalability and GPU optimization. It is subdivided into:

1. 2D preview Generator
2. 3D Renderer
3. Design Suggestion/Generation Engine
4. Generative Canvas Enhancer

2.2.4 Module 4: Data Management (MongoDB)

This module is responsible for storing all persistent data with clear schema separation. Each collection maps to a domain entity and supports fast querying and indexing.

1. Collection Creation
2. Collection Management
3. Ensuring Security
4. Creating interfaces and configurations

2.2.5 Module 5: Financial Operations (Payment Gateway)

This module handles all monetary flows with transparency and automation. This module ensures secure transactions and equitable revenue sharing.

1. Split Payments
2. Designer Payouts
3. Manufacturer Payouts
4. Premium Billing Plan

2.3 Tools and Technologies

The envisioned platform will be built using the MERN stack—MongoDB, Express.js, React, and Node.js—providing a robust and scalable foundation for web development. AI modules, developed in Python and trained via Google Colab, will offer intelligent design assistance, personalization, and predictive analytics. PyFlask will serve as the lightweight framework for deploying these models. GitHub will be used for version control, and Visual Studio Code will be the primary IDE for development. Tool/Technologies may change during execution phase if needed

2.4 Work Division

For each module and respective feature, the responsibility of each team member is given in the following table as a decimal showing the module number and feature number(e.g. x.y shows x module and y feature, module 1 sub modules are considered features here):

Table 2.1: Table 1

Name	Registration	Responsibility/ Module / Feature
Aisha Adil	22i-2718	1.2, 3.3, 4.1 for Designers,
Muhammad Uzair Choudhary	22i-2681	1.1, 1.2, 1.4, 3.1, 3.2
Anishra Ashfaq	22i-2614	1.1, 1.2, 1.4, 3.4

The unmentioned modules will be developed in collaboration along with testing and integration.

2.5 TimeLine

The timeline of our projects development life cycle is as follows:

Table 2.2: Table 2

Iteration#	Time frame	Tasks/Modules
01	Aug-Sep	AI module
02	Oct-Nov	Frontend (2 submodules) + integration of AI module
03	Dec-Jan	Frontend (2 submodules) + Data Management
04	Feb-Mar	Core Business Logic
05	April-May	Financial Operation, Testing and Integrations

Bibliography